

GAMING FPS PERFORMANCE ANALYSIS

A Power BI Dashboard on Hardware Performance

Capstone Project Report | Tool: Microsoft Power BI

11 CPU Models	22 GPU Models	30 Games	501 Highest FPS	3 Resolutions
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1. Introduction

Every gamer wants smooth gameplay, and that depends on FPS (frames per second). FPS is affected by many things: the GPU, the CPU, how much RAM is installed, the screen resolution, and settings like ray tracing. With so many hardware combinations available, it is hard to know which setup actually gives the best performance.

For this capstone project, I built a Power BI dashboard using gaming performance data. It covers 22 GPU models, 11 CPU models, 30 games, and 3 screen resolutions. The goal was to turn raw benchmark numbers into simple charts that show which hardware gives the best FPS, and under what conditions.

2. Challenge the Company Was Facing

A gaming hardware seller or tech review company deals with huge amounts of benchmark data, but without a clear report, they run into problems:

- Customers don't know which GPU or CPU actually gives the best FPS for their budget.
- It is hard to compare GPU brands like NVIDIA, AMD, and Intel fairly.
- Nobody has a clear answer on how much ray tracing slows down performance.
- There is no easy way to see which games are the most demanding on hardware.
- Customers have no simple guide showing which hardware combo crosses 100+ FPS.

3. Data Preprocessing Challenges I Faced

Before building the dashboard, I had to clean the raw benchmark data. This was not easy, since the data had several issues:

- **Duplicate entries:** Some GPU models had repeated test results. I had to find and remove these duplicates so the FPS averages stayed accurate.
- **Missing values:** A few rows were missing FPS or benchmark scores. I filled or removed these carefully so they would not affect the averages.
- **Inconsistent naming:** GPU and CPU model names were written in different ways (e.g. "RTX4070Ti" vs "RTX 4070 Ti"). I had to standardise all names before grouping them in charts.

- **Mixed formats:** Resolution was sometimes written as "1920x1080" and other times as "1080p". I converted everything into one consistent format.
- **Wrong data types:** Columns like RAM and VRAM were sometimes stored as text instead of numbers, so I had to convert them before doing any calculations.
- **Outliers:** A few benchmark scores looked unrealistically high or low. I checked these values and corrected or removed the incorrect ones.
- **Creating new categories:** There was no ready-made column to show which setups cross 100+ FPS, so I created a new category column myself to group hardware this way.

This cleaning step took real effort, but it was necessary. Wrong or messy data would have made every chart on the dashboard misleading.

4. The Approach I Decided to Take

Once the data was clean, I planned the dashboard around real questions a gamer or buyer would ask:

- Studied all columns: CPU, GPU, RAM, VRAM, resolution, ray tracing, FPS, and game engine.
- Grouped and cleaned the data so comparisons between brands and models were fair.
- Designed each chart to answer one clear question, like "Which GPU gives the highest FPS?"
- Added slicers for RAM and resolution so users can filter results themselves.
- Built a simple table showing hardware combos that give 100+ FPS.

5. Highlights of the Important Steps

- **Overview page:** Added KPI cards for total CPUs, GPUs, games, highest FPS, and resolution types, plus slicers for RAM and resolution.
- **FPS by GPU model:** Ranked all GPUs by highest FPS. The GeForce RTX 4080 and Radeon RX 7900 XTX came out on top.
- **Resolution comparison:** Found that 1080p is used the most (about 39%), followed by 1440p and then 4K, which needs the strongest hardware.
- **Ray tracing impact:** Compared FPS with ray tracing on and off. Turning ray tracing on clearly lowers FPS, dropping from about 130 to 100 on average.
- **Brand comparison:** Compared average FPS by brand. NVIDIA leads, followed by AMD, with Intel behind both.
- **Market share:** Checked GPU market share and found NVIDIA holds more than half the market, followed by AMD and then Intel.
- **RAM preference:** Found that most gamers use 16GB RAM, followed by 32GB, while 8GB and 64GB are less common.
- **Game engine and benchmark comparison:** Compared max FPS by game engine and benchmark scores by GPU to see which technology performs best.
- **Best hardware table:** Built a table listing CPU and GPU combinations that reach 100+ FPS, making it easy to pick a setup.

6. How This Will Help the Client Once Deployed

- **Easier buying decisions:** Customers can quickly see which GPU or CPU gives the best FPS for their budget and screen resolution.
- **Clear ray tracing guidance:** Buyers can decide if the visual quality of ray tracing is worth the drop in FPS, based on real numbers.
- **Smarter stock decisions:** The company can stock more of the GPUs and brands that customers prefer and perform best.
- **Better product recommendations:** The 100+ FPS table can be turned into a quick buying guide for customers who just want smooth gameplay.
- **Faster support and marketing:** Support teams and marketers can use the dashboard instead of digging through raw spreadsheets.

In short, this dashboard turns raw benchmark data into a simple tool that helps a hardware company guide customers, manage stock, and market smarter.

7. Conclusion

This dashboard gives a full picture of how hardware affects gaming performance, and a few clear patterns stand out. NVIDIA is the clear market leader, holding more than half of the GPU market and also topping the average FPS charts, which shows its popularity is backed by real performance rather than just brand name. AMD comes in second on both counts, while Intel remains a smaller player in both market share and raw FPS.

Ray tracing has a real cost: turning it on drops average FPS from about 130 to 100, so it is a genuine trade-off between visual quality and smoothness, not just a marketing feature. Most gamers still play at 1080p, and 16GB RAM is by far the most common setup, which shows that a hardware company's "sweet spot" products should be built around these two specs rather than high-end 4K, 64GB configurations. The data also confirms that higher VRAM generally lines up with higher average FPS, making VRAM a fair, simple way to guide budget-conscious buyers.

The best hardware table also shows that 100+ FPS is achievable across almost every CPU and GPU combination in the dataset, even with budget parts like the Core i3-12100, as long as the RAM and GPU are reasonably matched. This means smooth gameplay is possible at many price points, and customers do not always need to buy the most expensive hardware to get good performance.